

VACCINATION RASHES AND COMPLICATIONS.¹

By GEORGE PERNET, M.R.C.S. ENG., L.R.C.P. LOND.,

ASSISTANT TO THE SKIN DEPARTMENT, UNIVERSITY COLLEGE HOSPITAL; LATE PATHOLOGIST TO THE HOSPITAL FOR DISEASES OF THE SKIN, BLACKFRIARS, LONDON.

AT a time when we are passing through an epidemic of small-pox, and when large numbers of individuals are being vaccinated and revaccinated, I have thought it would be opportune and interesting to review shortly the subject of vaccination rashes and complications. In dealing with this matter I have no fear of supplying weapons to those who oppose the practice of vaccination. It is the function of a science of observation like medicine to record carefully and impartially all phenomena, normal or otherwise, that we may thereby increase the sum of our knowledge and the usefulness of our art. It is not my purpose in this paper to discuss the value of vaccination—that is a point, I take it, upon which we are all agreed. I would merely remark that the advancement of all knowledge is a process of slow and laborious evolution. I mention this because it is the stereotyped fashion in some quarters to make much of the fact that Jenner expressed the opinion that vaccination absolutely protected against small-pox. We are wiser now and know that vaccination in infancy does not confer complete immunity. It must not be forgotten, however, that small-pox itself does not absolutely protect an individual against a second attack, any more than one attack of herpes zoster excludes the possibility of a second, although in both diseases it is extremely rare for two attacks to occur. Louis XV. of France is said to have had variola at the age of 14 years, but that did not prevent him dying from a second attack at the age of 74.² Again, there can be no doubt that cases of syphilitic re-infection have occurred. As far as Jenner is concerned I may be permitted to say that his original contributions are models of lucidity and honesty of purpose and breathe the spirit of Harvey's noble "De Motu Cordis."

It will add, I think, to the clearness of my remarks if I adopt the following division of the subject, suggested originally by Dr. T. Colcott Fox,³ which is at once simple and natural, according with the manifestations as they present themselves clinically: (1) local abnormalities or irregularities in the development of the vaccine vesicles; (2) incidental exanthematic eruptions; (3) diseases inoculated with vaccinia at the time of the operation; (4) diseases of a septic nature which find a nidus in the wounds subsequently to the operation; and (5) diseases excited in subjects specially predisposed to the same. To which I will add—(6) accidental vaccination; (7) horse-pox or grease and sheep-pox in man; (8) effects of intercurrent diseases on vaccination and *vice versa*; and (9) vaccination as a mode of treatment.

1. Among the local abnormalities extra vesicles sometimes occur, due probably to the movements of the child and more points being inoculated than was intended. At other times bullæ may form instead of vesicles (vaccinia pemphigoides). In the case of calf lymph this arises as far as I can ascertain from impotent lymph. The bullæ leave no scars. In such an event vaccination should be done over again with good lymph. Jenner referred to this as "spurious" cow-pox and pointed out that it was not protective. He insisted on the necessity of re-inoculation with the proper virus.⁴ Again, vesicles may be retarded in their development or may not form at all. In connexion with the late appearance of vesicles curious instances of delay have been recorded: two months (Brierly⁵), one year (George

Harley⁶). But the most remarkable was a delay of 14 years, recorded by no less an authority than Sir Thomas Watson. The patient was a girl 14 years of age who had been vaccinated in infancy. In the course of an attack of influenza vesicles developed on the old vaccination areas and, what is more, an elder sister was revaccinated with the lymph of the aforesaid vesicles with positive results.⁷ There are also curious instances of revivification of old vaccination areas as a result of revaccination or a subsequent attempt at vaccination where the first had failed. In some cases, instead of drying up in the normal way, the vaccination areas may go on to ulceration with induration of the base and exudation, forming crusts accompanied by axillary adenitis. Such a condition might give rise to the erroneous diagnosis of syphilis. These ulcers have been called chancriform by the French. In one instance, Leloir and Hervieux investigated an epidemic of this kind, 37 out of 38 children being affected. The lymph was of human origin, but no specific symptoms developed.⁸ Gangrene has also been observed as a local complication. Diday has recorded the growth of hair as a result of vaccination forming a coronet round the areas.⁹

There has been some discussion in the journals with regard to relapse of vaccination.¹⁰ If by that is meant a formation of typical vesicles going through a normal course and following on a successful vaccination, I have never seen such a thing. But I have seen swelling up of one area (out of three) commencing a few days after the shedding of the scabs after successful vaccination. The purplish red lesion was of the diameter of a sixpence, flatly raised at first, but becoming convex. It felt soft in the centre, but on making a puncture no fluid escaped, the contents being jelly-like. Under treatment (euphen and blue wool and collodion) it gradually flattened down leaving (three months later) a slightly raised scar.¹¹ There was no axillary adenitis. I could only put this complication down to the staphylococcus. The patient had probably inoculated the area with her nails, and this was the more likely as she had a few acne pustules about the chin. In a case under the care of Mr. F. Gower Gardner of Warwick, small red spots appeared after successful revaccination and gradually enlarged to the size of the vaccinated areas.¹² Hypertrophic scars may develop on vaccinated areas.¹³ Such a condition is sometimes recorded as keloid, but it is important to bear in mind the differences between true keloid and hypertrophic scar.

2. As to the incidental rashes some occur early. They are usually urticarial or erythematous and are in most cases evanescent, but they may also be vesicular and even bullous. As to the last named in infants and young children, the possibility of impetigo contagiosa bullosa must not be lost sight of, especially when there are only a few bullæ irregularly scattered about one limb or the face. A fairly common condition is the so-called vaccine lichen, consisting of a number of pin-point, red, conical papules, which come out in crops and may affect many parts of the body. According to Dr. H. Radcliffe Crocker, the rash comes out from the fourth to the eighteenth day, usually about the eighth.¹⁴ The character of the eruption is in some cases papulo-vesicular, exceptionally it is pustular. This kind of rash had apparently been noted by Jenner. More rarely still the groups of small vesicles may run together, forming bullæ, but this is a rare occurrence. As a rule it soon subsides but in other cases it may be very persistent. The same may be said of urticaria following vaccination. The fact is that urticaria once started is sometimes difficult to quell. Erythema multiforme may also ensue, no doubt as a result of a toxæmia, the rash appearing as papules about the backs of the hands and face; or it may take the form of an erythema iris¹⁵ or erythema bullosum. These conditions may occur when the vaccination areas are nearly or quite healed. Impetigo contagiosa is not uncommon in neglected children and many parts of the body may be involved by

¹ A paper read before the Medical Society of University College, London, on March 19th, 1902.

² Trousseau: Clinique Médicale de l'Hôtel-Dieu, Paris, 1865, second edition, vol. i., p. 34.

³ Medical Times, 1877, vol. ii., p. 621, cited by Poole, op. cit.

⁴ Jenner: On the Varieties and Modifications of the Vaccine Pustules occasioned by a Herpetic State of the Skin. (Originally published in the Medical and Physical Journal, August, 1804, No. 66. Also printed separately, Cheltenham, 1806. This paper is not included in Crookshank's History and Pathology of Vaccination, vol. ii., which contains many interesting reprints of original contributions by Jenner and others.) The above separately published paper is in the British Museum library.

⁵ Medical Times, 1862, p. 442. Cited by Poole in Vaccination Eruptions, 1893, p. 24.

⁶ Medical Times, 1881, vol. ii., p. 572. Cited by Poole, op. cit.

⁷ Vide note 3.

⁸ Cited by Delobel and Cozette in Vaccine et Vaccination, p. 67. See also Kapusi-Besnier, second French edition, vol. i., p. 325.

⁹ Radcliffe Crocker: Diseases of the Skin, second edition, p. 324.

¹⁰ Brit. Med. Jour., vol. i., 1902, p. 496; also vol. i., 1902, p. 568.

¹¹ Private Case Book, C, p. 208.

¹² Letter to Author, March 3rd, 1902.

¹³ Ehlers: Dansk Dermatologisk Selskabs Forhandling, 1899, p. 41. Also Pernet: British Journal of Dermatology, vol. xiv.

¹⁴ Op. cit., supra, p. 323.

¹⁵ Norman Walker: Brit. Med. Jour., vol. i., 1901, p. 1201.